USN | | | | | | | | | | | [22BS2MA01]

SECOND SEMESTER B.Tech. EXAMINATION, JUNE - 2023

PROBABILITY AND VECTOR SPACES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

PART - A

05 X 02=10

1.	**a.**	A fair six-sided die is rolled three times. Find the probability of getting at least one 6 in the three rolls.	L-1	CO1	**02**
	b.	List out basic components of random experiment.	L-1	CO1	**02**
	c.	Express $\begin{bmatrix}2\\4\end{bmatrix}$ as a linear combination of $u_1 = \begin{bmatrix}2\\1\end{bmatrix}, u_2 = \begin{bmatrix}2\\-2\end{bmatrix}$.	L-2	CO3	**02**
	d.	Find the matrix for $T: V_2 \to V_3$ is a linear transformation such that $T(x_1, x_2) = (x_1 + x_2, 2x_1 - x_2, 7x_2)$.	L-2	CO3	**02**
	e.	Define right and left singular vectors of a matrix.	L-1	CO1	**02**
		PART - B			**6 X 5=30**
2.	**a.**	The two cards are drawn in succession from a pack of 52 cards. Find the probability that first is king and the second is queen if the first card is (a) replaced (b) without replaced.	L-3	CO1	**05**
		OR			
	b.	If a die is rolled twice and the first number is 3, find the probability that the sum of numbers is 8.	L-3	CO1	**05**
3.	**a.**	20 wrist watches in a box of 100 are defective. If 10 watches are selected at random. Find the probability that (i) 10 are defective (ii) atleast one watch is defective.	L-3	CO2	**05**
		OR			
	b.	The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that random call made from this booth (i) ends in less than 5 minutes (ii) ends between 5 and 10 minutes.	L-3	CO2	**05**
4.	**a.**	Show that the subset $w = \{(x_1, x_2, x_3) \mid x_1 + x_2 + x_3 = 0\}$ of the vector space $V_3(R)$ is a subspace of $V_3(R)$.	L-3	CO3	**05**
		OR			
	b.	Let $S = \{(1,-3,2), (2,4,1), (1,1,1)\}$ be a subset of $V_3(R)$. Check whether the vector $(3,-7,6)$ is in $L(S)$ or not.	L-3	CO3	**05**
5.	**a.**	Find the linear transformation $f: R^2 \to R^2$ such that $f(1, 1) = (0, 1)$ *and* $f(-1, 1) = (3, 2)$.	L-3	CO4	**05**
		OR			
	b.	Show that $\{v_1, v_2, v_3\}$ is an orthonormal set of R^3, where	L-3	CO4	**05**

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		$v_1 = \frac{1}{\sqrt{11}}\begin{bmatrix}3\\1\\1\end{bmatrix}, v_2 = \frac{1}{\sqrt{6}}\begin{bmatrix}-1\\2\\1\end{bmatrix}$ and $v_3 = \frac{1}{\sqrt{66}}\begin{bmatrix}-1\\-4\\7\end{bmatrix}$.			
6.	**a.**	Let $A = \begin{pmatrix}1 & 6\\5 & 2\end{pmatrix}$, $u = \begin{pmatrix}6\\-5\end{pmatrix}$ and $v = \begin{pmatrix}3\\-2\end{pmatrix}$. Are u and v eigen vectors of A?	L-3	CO6	**05**
		OR			
	b.	Find Eigen Values and Eigen Vectors of the matrix $A = \begin{bmatrix}-2 & 2\\5 & 1\end{bmatrix}$.	L-3	CO6	**05**
7.	**a.**	Describe python code to find $i)\ P(X = 5),\ ii)\ P(X \geq 7),\ iii)\ P(1 \leq x \leq 6)$ using binomial distribution where mean is 3 and variance is 9/4.	L-3	CO2	**05**
		OR			
	b.	Suppose 36% of families own a dog, 30% of families own a cat, and 22% of the families that have a dog also have a cat. A family is chosen at random and found to have a cat. Write a python code to find the probability that they also own a dog?	L-3	CO1	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	Two Aero planes bomb a target in succession. The probability of each correctly scoring a hit is 0.3 and 0.2, respectively. The second will bomb only if the first misses the target. Find the probability that **i.** The target is hit **ii.** Both fails to score hits	L-3	CO1	**10**
		OR			
	b.	Three major parties A, B and C are contending for power in the elections of a state and the chance of their winning the election is in the ratio 1:3:5. The parties A, B and C respectively have probabilities of banning the online lottery 2/3, 1/3 and 3/5. What is the probability that there will be a ban on the online lottery in the state? What is the probability that the ban is from the party C?	L-3	CO1	**10**
9.	**a.**	A random variable X has the following probability function for various values of x.	L-3	CO2	**10**

X	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

i. Find k

ii. Find mean, variance and standard deviation

iii. $p(x < 6),\ p(x \geq 6)\ and\ p(3 < x \leq 6)$

		OR			
	b.	In a normal distribution, if 31% of items are under 45 and 8% of items are over 64, then find the mean and standard deviation of this distribution. [Given $\phi(0.5) = 0.19$ and $\phi(1.4) = 0.42$]	L-3	CO2	**10**
10.	**a.**	Find the four fundamental subspace of the matrix A where $A = \begin{bmatrix}1 & 3 & 2\\2 & 6 & 4\\0 & 1 & 1\end{bmatrix}$.	L-3	CO3	**10**

		OR			
	b.	Prove that the set V_n of all ordered $n-$tuples of real numbers is a vector space.	L-3	CO3	**10**
11.	**a.**	Show that $\{u_1,\ u_2,\ u_3\}$ is an orthogonal set where $u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ $and\ u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$. Further express $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as linear combination of vector S.	L-3	CO4	**10**
		OR			
	b.	Consider the basis $S = \{u_1, u_2, u_3\}$ for R^3 where $u_1 = (1,1,1)$, $u_2 = (-1,0,-1)$ and $u_3 = (-1,2,3)$. Use the Gram- Schmidt process to transform S to orthonormal basis for R^3.	L-3	CO4	**10**
12.	**a.**	In the manufacturer of product X, the amount of the compound beta present in the product is controlled by the amount of independent alpha used in the process. In manufacturing the gallon of X, the amount of alpha used, and the amount of beta present are recorded. The following data were obtained	L-3	CO5	**10**

Alpha used (Ounces/ gallon)	3	4	5	6	7	8	9	10	11	12
Beta Present (Ounces/ gallon)	4.5	5.5	5.7	6.6	7.0	7.7	8.5	8.7	9.5	9.7

i. Find an equation of the least square line for the given data.

ii. Use the equation obtained to predict the number of ounces of beta present in a gallon of product X if 30 ounces of alpha are used per gallon.

		OR			
	b.	What is the necessary condition for a matrix to be orthogonally diagonalizable? Further orthogonally diagonalize the matrix $\begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$.	L-3	CO6	**10**
13.	**a.**	Find left singular vectors of the matrix $\begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$.	L-3	CO6	**10**
		OR			
	b.	Describe Python code to show that the set of vectors (1,2,1), (3,4,7) and (3,6,3) are linearly dependent.	L-3	CO3	**10**